

Neptune

Overview

Neptune is the eighth and farthest known planet from the Sun, orbiting at an average distance of 4.495 billion km — 30 times farther from the Sun than Earth. At this distance, sunlight reaching Neptune is about 900 times weaker than at Earth, and the planet receives so little warmth from the Sun that its upper atmosphere reaches -218 °C. Neptune takes 164.8 Earth years to complete one orbit, so since its discovery in 1846, it has only just completed its first full orbit observed by humans. With a diameter of 49,244 km — about four times that of Earth — Neptune is the fourth-largest planet by diameter and third-largest by mass. Like Uranus, Neptune is classified as an ice giant, with an interior rich in water, ammonia, and methane ices surrounding a rocky core.

Discovery

Neptune holds the distinction of being the first planet whose existence was mathematically predicted before it was directly observed. By the 1840s, astronomers had noticed that the orbit of Uranus deviated slightly from predictions based on known gravitational influences, suggesting an undiscovered planet was tugging on it. French mathematician Urbain Le Verrier and British mathematician John Couch Adams independently calculated where this unknown planet should be. On 23 September 1846, German astronomer Johann Galle, using Le Verrier's predictions, pointed his telescope at the calculated position and found Neptune within 1 degree of the predicted location — a triumph of celestial mechanics. Neptune's largest moon, Triton, was discovered just 17 days later by British astronomer William Lassell.

Atmosphere and Winds

Neptune has the most violent weather of any planet in the Solar System, with wind speeds exceeding 2,100 km/h — faster than the speed of sound on Earth and the highest recorded in the Solar System. These extreme winds are puzzling given that Neptune receives so little solar energy; they are thought to be driven primarily by Neptune's internal heat source. Neptune radiates about 2.61 times more energy than it receives from the Sun, indicating significant primordial heat still escaping from its interior — in sharp contrast to Uranus, which radiates almost no internal heat. Voyager 2 observed a large storm system called the Great Dark Spot in 1989, comparable in size to Earth, but when the Hubble Space Telescope looked at Neptune in 1994 the original spot had disappeared and new dark spots had formed elsewhere, suggesting that Neptunian storms are far more transient than Jupiter's Great Red Spot. Neptune's vivid blue colour is more saturated than Uranus's despite similar methane content, suggesting an additional unknown chromophore compound is present.

Triton and Other Moons

Neptune has 16 known moons. The largest, Triton, is unique among large moons in the Solar System because it orbits Neptune in a retrograde direction — opposite to Neptune's own rotation — at an inclination of 157° to Neptune's equator. This retrograde orbit is strong evidence that Triton did not form in place around Neptune but was captured from the Kuiper Belt, making it the largest known captured body in the Solar System. Triton is geologically active, with nitrogen geysers erupting from its surface and reaching up to 8 km into its thin atmosphere, observed by Voyager 2 in 1989. The tidal forces of Neptune's gravity are slowly causing Triton's orbit to decay; in approximately 3.6 billion years, Triton will pass within Neptune's Roche limit and be torn apart by tidal forces, creating a new ring system that could rival Saturn's in splendour. Nereid, discovered in 1949, has the most eccentric orbit of any moon in the Solar System, swinging from 1.4 million km to 9.7 million km from Neptune.

Exploration

Neptune has been visited by only one spacecraft in history: NASA's Voyager 2, which flew past on 25 August 1989, providing the only close-up images and measurements ever taken of the planet and its moons. During the flyby, Voyager 2 discovered six new moons, characterised the ring system, measured wind speeds, and photographed Triton's geysers and complex terrain. Because of Neptune's enormous distance, a spacecraft launched today would take approximately 12 years to reach it even using gravitational assists. ESA and NASA are studying concepts for a dedicated Neptune orbiter mission, sometimes called the Neptune Odyssey concept, which would orbit Neptune for at least four years and send a probe into its atmosphere and possibly land on Triton. A potential launch window in the early 2030s using a Jupiter gravity assist would reduce travel time to about 12 years, potentially arriving in the mid-2040s.